Those who had the technology and engines were the USSR, the US, France, Japan and China. Even if one of them were to sell us, making our own engine and stage, it would take us several years of training under the guidance of the technology seller. The perennial truth in rocket science: getting all the blueprints and even a few pieces of finished engines do not enable one to put together a rocket engine. You need the 'know-why' beyond the know-how. So the only option would be to strike a deal with a country that has the technology to buy a few engines and also get a technology transfer with long-term collaboration for development.

While discussions were on to speed up cryogenic acquisition, that day in 1975 crossed my mind, when I was at the Paris, at Le Bourget Airport for the Aerospace Exhibition. Among the rocket engines on display was a Russian engine labelled RD-100. In one glance I knew it was a cryogenic engine, but something told me it was not RD-100.

'They've got the label wrong,' I had told my accompanying scientist Mohana Prasad. Just when we were walking away after searching for a Russian scientist who could clarify our doubt, an elderly man approached us. We had seen him at a distance; he appeared to be more of a helper. Even Russian scientists were not very conversant in English, so we had not bothered to consult this man in wrinkled clothes.

'Now, gentleman, what makes you think it's not an RD-100?' he said.

Prasad and I looked at each other in surprise before I replied.

'Because it doesn't look like one.'

'It is indeed an RD-100,' he said in English.

I disagreed.

'Tell me why?' he insisted.

'An RD-100 has a different type of a cooling passage,' I said.

The Russian had an expression which I could not ascertain as confused or amused.

'Can we discuss this over dinner?' he said.

Over dinner that night, the Russian scientist explained that it was not an RD-100 engine. It was a cryogenic engine called RD-56, also called the